Q **August 13, 2026** SEAN LI

Tossing the Wok

[7 pts]

Assumptions

1. The accelerations are prescribed kinematically — think of them as constraints imposed by the pan, not as net forces acting on a free body. Do NOT add g to the net acceleration when $t < t_0$.

Dr. Wen is learning to *toss the wok* — the classical technique of jerking a pan sharply to flip its contents. Model the pancake Ω as a uniform rigid disk of radius R and planar mass density ρ, initially lying in the xy-plane with its center of mass at the origin.
At $t = 0$, two constant accelerations are applied for $[0, t_0]$:

1. Angular acceleration α about an in-plane axis through the center of mass (the "flipping axis").

2. Linear acceleration $\vec{\mathbf{a}}$ of the CM, vertical component a.

At $t = t_0$ the forces cease. An n-**toss** is *successful* if at some $t' > t_0$ the pancake returns exactly to its original spatial region, and the rigid motion between the two configurations is a pure rotation by $n\pi$ about the flipping axis.
Dr. Wen cooks in vacuum. Do not consider air resistance.

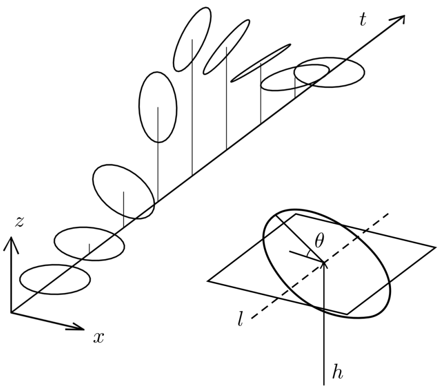

Figure 1: The pancake at different t during a 1-toss. θ is the rotation about the flipping axis ℓ ($\ddot{\theta} = \alpha$ for $t < t_0$), and h the CM height ($\ddot{h} = a$ for $t < t_0$).

Questions.

Q1. Toss Constraint. Formulate an equation that relates n, α, a, and t_0 for any $n > 0$.

Q2. Toss Energy Ladder. Derive the work W done on the pancake as an expression containing n, α, a and other physical constants, and show that $W \propto n$.

Q3. Cracked Pancake.

While tossing, Dr. Wen's pancake cracks exactly along the flipping axis, splitting it into two equal half-disks. He hastily glues them back together with egg yolk, which can provide a total binding force of at most F_y normal to the crack.
During the forced phase ($t < t_0$), the rotation accelerates, creating an outward centrifugal tendency on each half-disk. If the required centripetal force exceeds F_y, the halves separate and the toss is ruined. Given n (and the toss constraint from Q1), find the minimum F_y as an expression of n, a, α, m and other physical constants needed to prevent splitting.